



PABNA UNIVERSITY OF SCIENCE AND TECHNOLOGY

LAB REPORT

Course Code: ICE-3204

Course Title: Antenna Engineering Sessional

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1. Study and Simulation of the 3D Radiation Pattern of a Dipole Antenna

Objective

To study and simulate the radiation pattern of a dipole antenna.

Theory

A dipole antenna is one of the most basic practical antennas. Its radiation is maximum in the direction perpendicular to the antenna axis and minimum along the axis. In a simplified form, the radiation pattern can be represented by

$$R(\theta) = |\sin(\theta)|$$

where θ is the observation angle. The pattern is symmetrical and forms a figure-eight shape in polar form. This is a fundamental antenna pattern used for studying directional behavior.

Code

```
import numpy as np
import matplotlib.pyplot as plt

# Angular values
theta = np.linspace(0, np.pi, 100)
phi = np.linspace(0, 2*np.pi, 100)
theta, phi = np.meshgrid(theta, phi)

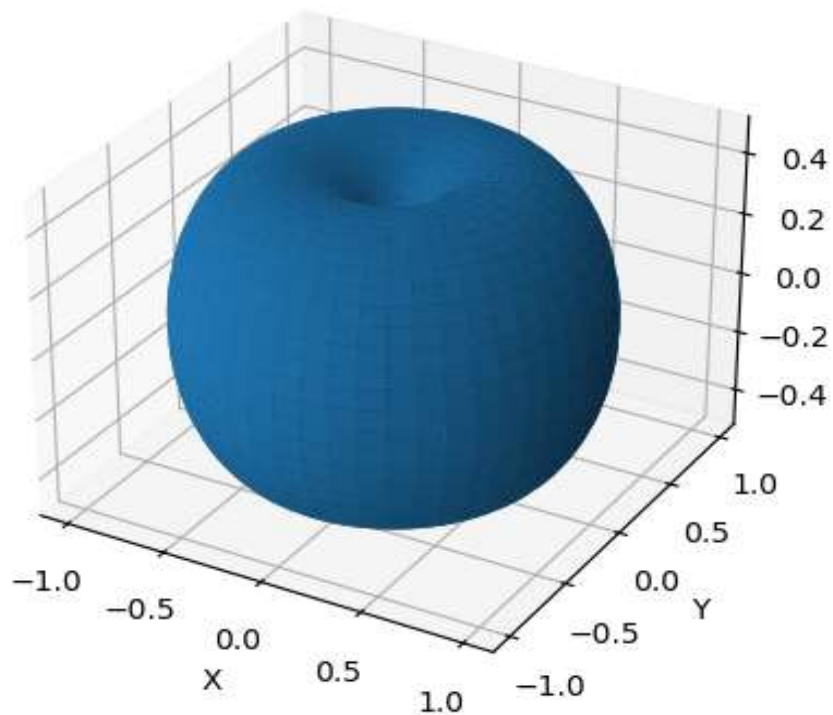
# Radiation pattern formula for half-wave dipole
R = np.abs(np.sin(theta))
# Convert to Cartesian coordinates
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

# Plotting 3D pattern
fig = plt.figure()
```

```
ax = fig.add_subplot(projection='3d')
ax.plot_surface(X, Y, Z)
ax.set_title("3D Radiation Pattern of Dipole Antenna")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
plt.show()
```

Output

3D Radiation Pattern of Dipole Antenna



Result & Discussion

This lab demonstrates the basic directional property of a dipole antenna. The use of $\text{np.sin}(\theta)$ models the angular dependence of radiation. The $\text{abs}()$ function ensures that the pattern remains positive because radiation intensity cannot be negative. The polar plot is appropriate because antenna patterns are angle-based. This result is important as a foundation for understanding practical antenna directivity.

2. Study and Simulation of the 3D Radiation Pattern of an Isotropic Antenna

Objective

To plot the ideal isotropic antenna radiation pattern and understand its uniform radiation behavior.

Theory

An isotropic antenna is a theoretical reference antenna that radiates equally in all directions. In practice, such an antenna does not exist, but it is widely used as a benchmark for gain comparison. Its normalized radiation pattern is constant:

$$R(\theta) = 1$$

In 2D, this constant value produces a circle in a polar plot.

Code

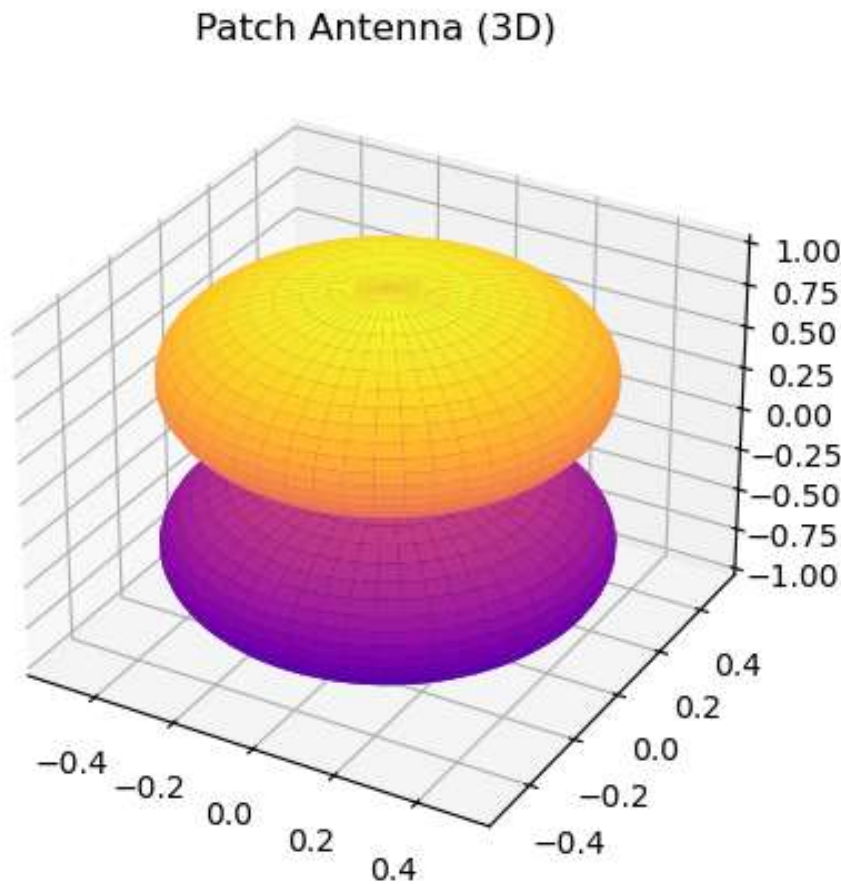
```
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 200)
phi = np.linspace(0, 2*np.pi, 200)
theta, phi = np.meshgrid(theta, phi)
R = np.abs(np.cos(theta))

X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="plasma")
ax.set_title("Patch Antenna (3D)")
plt.show()
```

Output



Result & Discussion

This lab represents an ideal antenna model used for reference in antenna engineering. The function `np.ones_like(theta)` generates a constant radius for every angle, resulting in a circle. This pattern helps define antenna gain in dBi, where the isotropic antenna is the zero-reference point. Although impossible to realize physically, the isotropic model is essential in analysis and comparison.

3. Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective

To plot the simplified radiation pattern of a horn antenna and observe its directional behavior.

Theory

A horn antenna is a highly directional microwave antenna. It is commonly used in radar, satellite communication, and measurement systems. A simplified radiation model is often expressed as

$$R(\theta) = |\cos(\theta)|^n$$

where n controls the beam sharpness. Larger values of n produce narrower beams and higher directivity.

Code

```
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 250)
phi = np.linspace(0, 2*np.pi, 250)
theta, phi = np.meshgrid(theta, phi)

n = 6
R = np.abs(np.cos(theta))**n

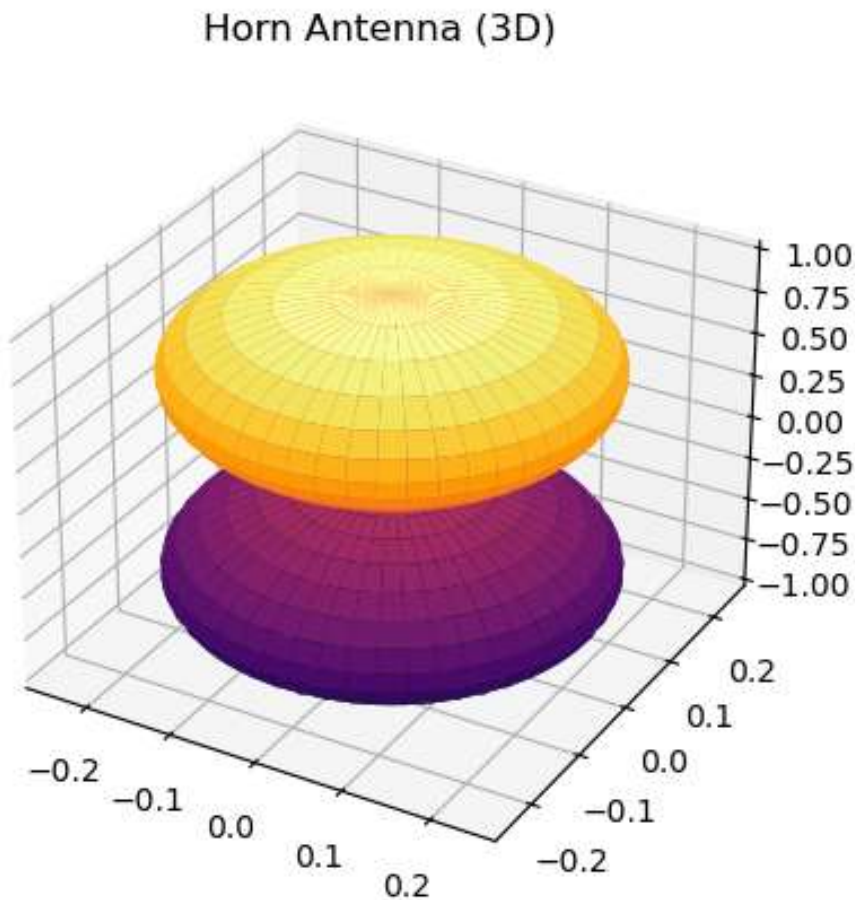
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

ax = plt.figure().add_subplot(projection='3d')
```



```
ax.plot_surface(X, Y, Z, cmap="inferno")
ax.set_title("Horn Antenna (3D)")
plt.show()
```

Output



Result & Discussion

This lab highlights the main property of a horn antenna: high directivity. The cosine power model is a useful approximation for visualization. Increasing n makes the beam more focused. The angular range is limited to -90° to $+90^\circ$ because horn antennas primarily radiate forward. This pattern is much more directional than the isotropic and dipole patterns.

4. Study and Simulation of the 2D Radiation Pattern of a Linear Antenna

Objective

To compute and plot the radiation pattern of a linear antenna array using the array factor.

Theory

A linear array consists of multiple antenna elements placed in a straight line. The total radiation pattern depends on the interference of all elements and is described by the array factor. For an array of identical elements, the normalized array factor is

$$AF(\theta) = \left| \frac{\sin\left(\frac{Nkd \cos \theta}{2}\right)}{N \sin\left(\frac{kd \cos \theta}{2}\right)} \right|$$

Code

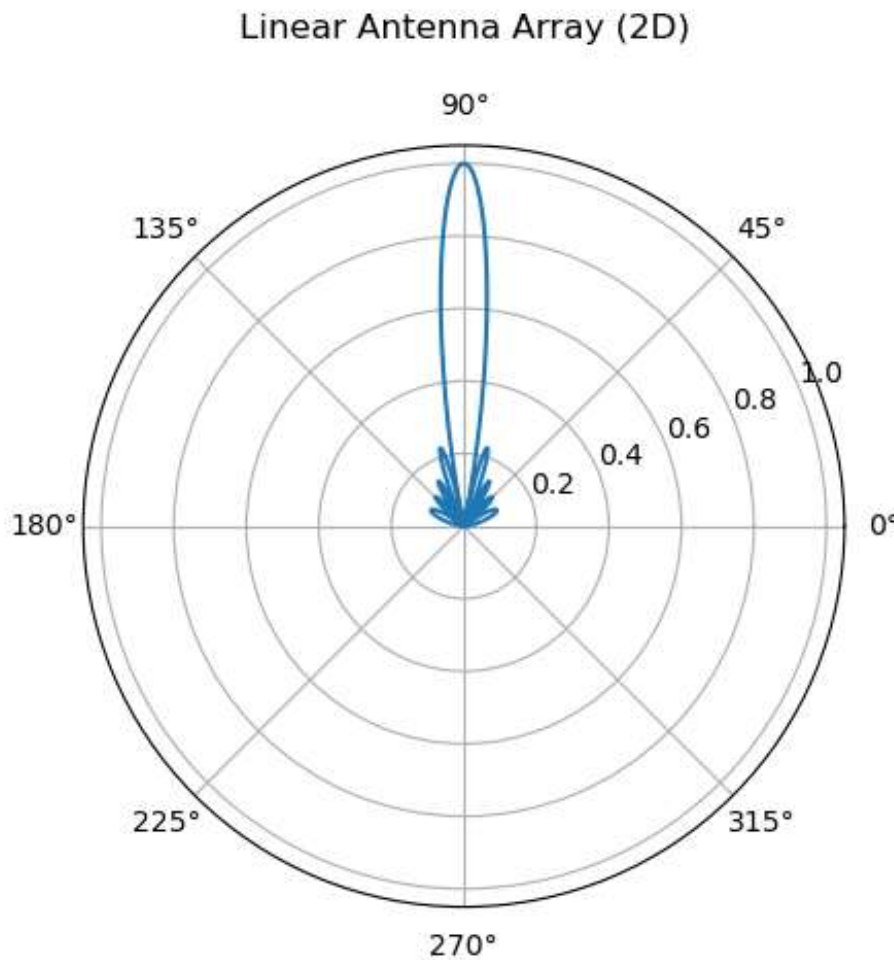
```
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 800)
N = 10
d = 0.5
k = 2*np.pi

num = np.sin(N * k * d * np.cos(theta) / 2)
den = N * np.sin(k * d * np.cos(theta) / 2)
AF = np.abs(num / den)

ax = plt.subplot(111, projection="polar")
ax.plot(theta, AF)
ax.set_title("Linear Antenna Array (2D)\n")
plt.show()
```

Output



Result & Discussion

This lab explains the principle of array antennas. The array factor is used because the combined radiation depends on phase differences between elements. When the number of elements increases, the beam becomes narrower and the directivity increases. The element spacing also affects the appearance of side lobes and grating lobes. This experiment is important for understanding beamforming and phased-array systems.

5. Study and Simulation of the 2D Radiation Pattern of a Broadside 3 Element Array Antenna

Objective

To plot the radiation pattern of a three-element broadside antenna array.

Theory

In a broadside array, all elements are fed in phase, and the maximum radiation occurs perpendicular to the axis of the array.

For a linear array of N elements, the array factor can be written as

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jnkd \cos \theta}$$

Code

```
import numpy as np
import matplotlib.pyplot as plt

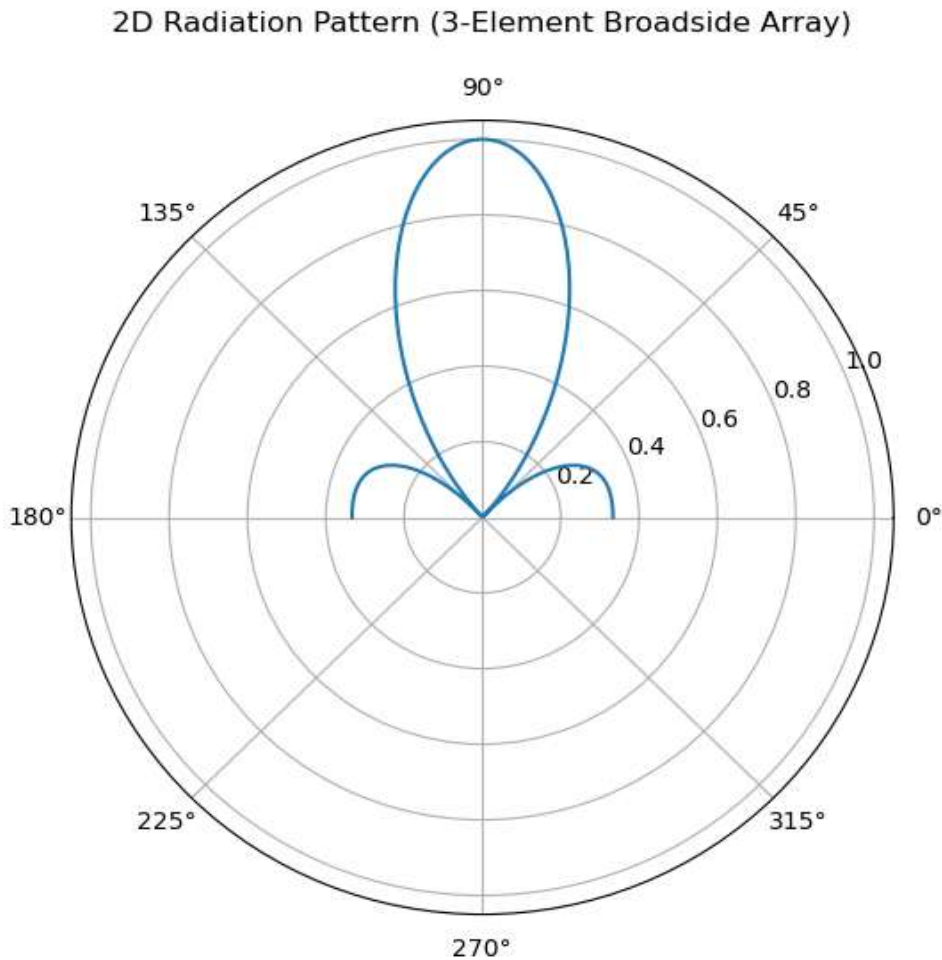
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 3 # number of elements (changed)

theta = np.linspace(0, np.pi, 1000)
AF = np.zeros_like(theta, dtype=complex)
for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
```

```
ax.set_title("2D Radiation Pattern (3-Element Broadside Array)\n")  
plt.show()
```

Output



Result & Discussion

This lab shows how multiple antenna elements can be combined to achieve directional radiation. Since the elements are in phase, the fields add constructively in the broadside direction. The result is a symmetric radiation pattern with a strong central main lobe. Broadside arrays are widely used in radar and wireless systems because they provide high gain in a desired direction.

6. Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Phase ($d = \lambda/2$)

Objective

To plot the far-field radiation pattern of two isotropic sources having equal amplitude and the same phase.

Theory

When two identical sources are separated by a distance and excited with equal amplitude and equal phase, the resulting radiation pattern is obtained by interference. For two sources, the array factor is

$$AF(\theta) = 2 \cos \left(\frac{kd}{2} \cos \theta \right)$$

With $d = \lambda/2$, the resulting pattern is bidirectional.

Code

```
import numpy as np
import matplotlib.pyplot as plt

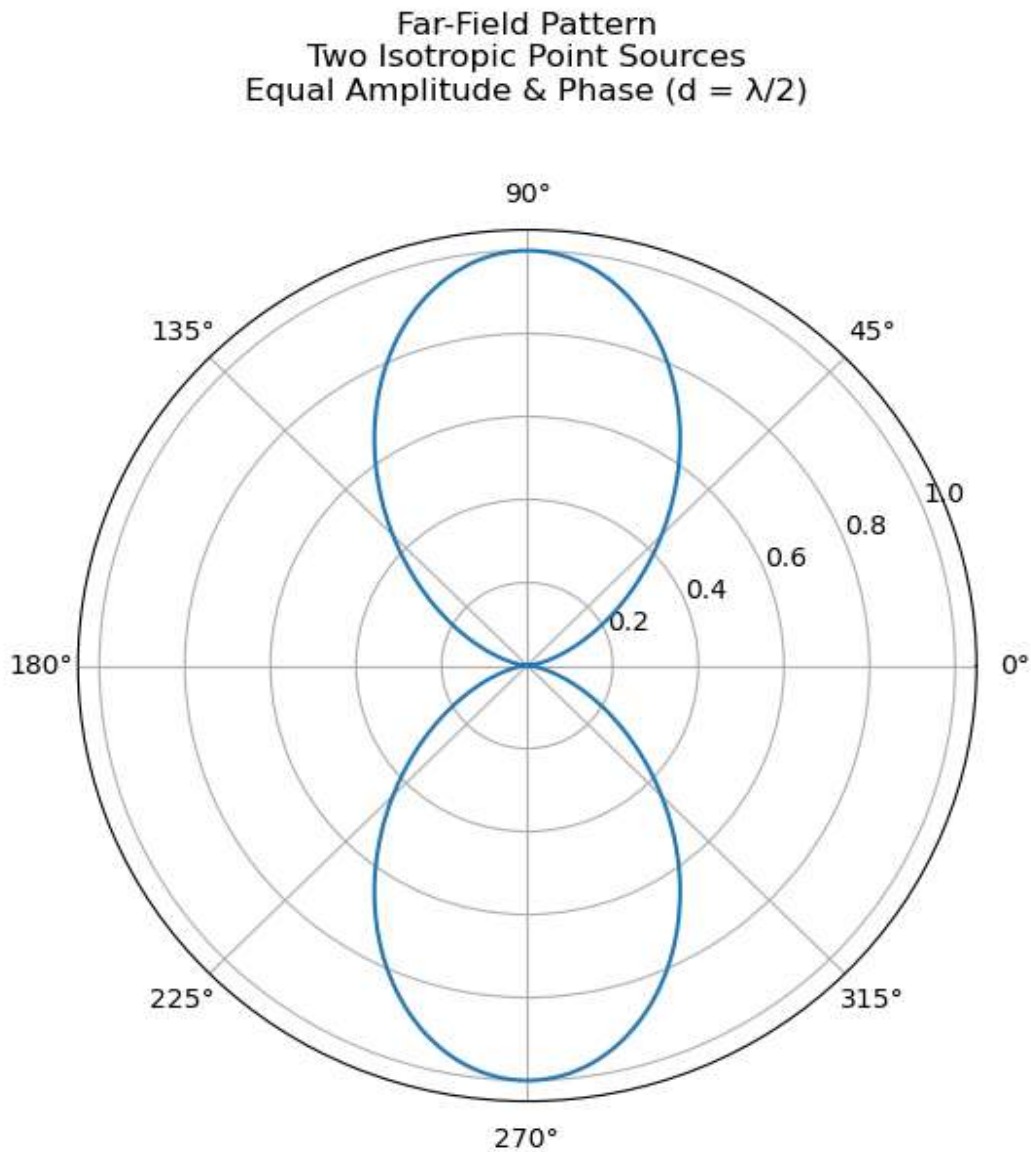
lam = 1.0          # wavelength (normalized)
d = lam / 2        # element spacing
k = 2 * np.pi / lam # wave number

theta = np.linspace(0, 2*np.pi, 1000)
AF = 2 * np.cos((k * d / 2) * np.cos(theta))

AF_norm = np.abs(AF) / np.max(np.abs(AF))

plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nEqual\nAmplitude & Phase (d =  $\lambda/2$ )\n\n")
plt.show()
```

Output



Result & Discussion

This lab is important because it introduces the simplest antenna array behavior. With equal phase excitation, the fields add constructively at broadside and destructively along the axis. The result resembles a dipole-like pattern. This experiment provides the foundation for understanding array interference and the role of spacing.

7. Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Opposite Phase ($d = \lambda/2$)

Objective

To plot the far-field radiation pattern of two isotropic sources with equal amplitude and opposite phase.

Theory

If two identical sources are excited with a phase difference of 180° , the resulting pattern changes significantly. The array factor becomes

$$AF(\theta) = 2 \sin \left(\frac{kd}{2} \cos \theta \right)$$

Code

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1.0          # wavelength (normalized)
d = lam / 2        # spacing
k = 2 * np.pi / lam # wave number

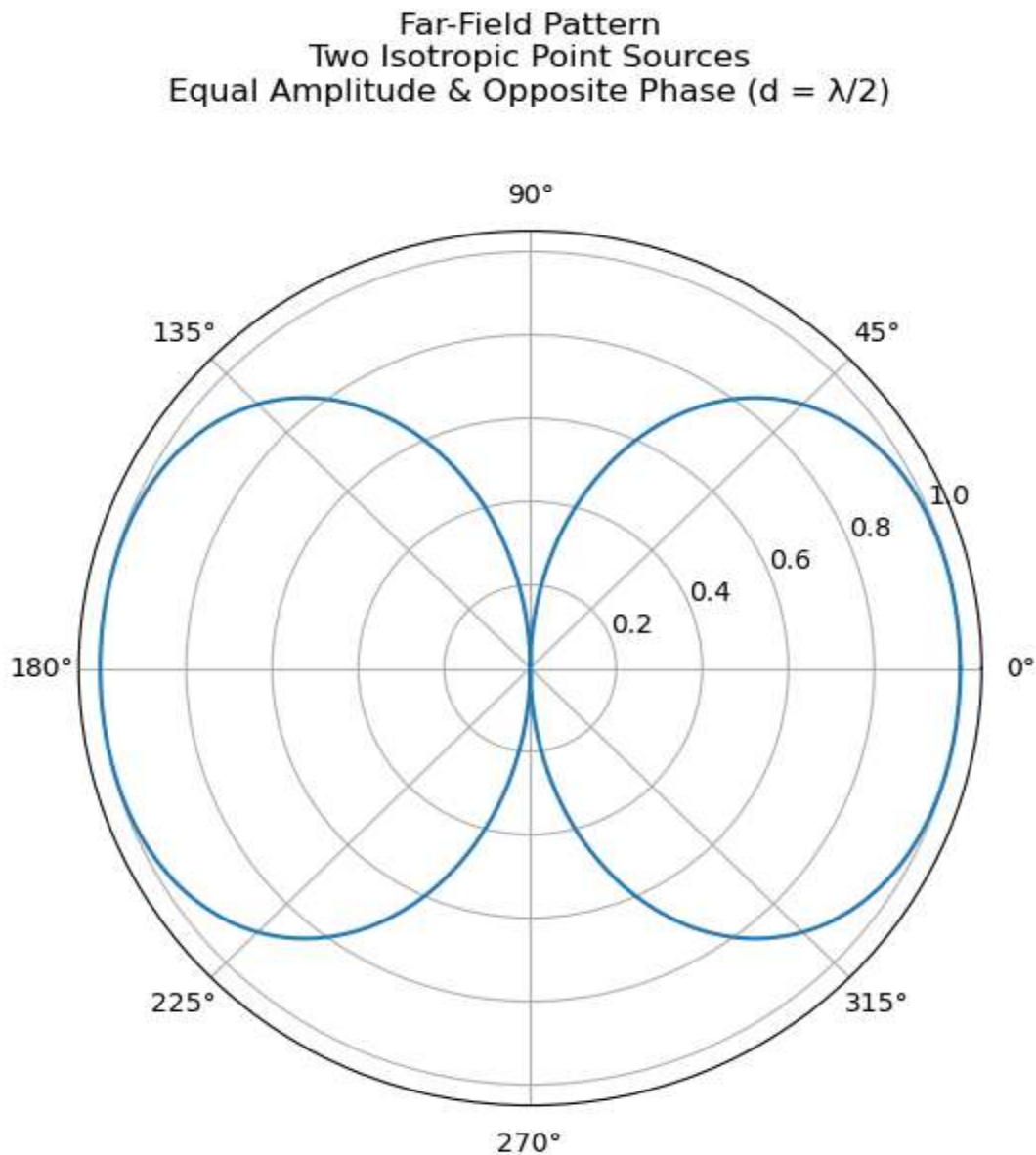
theta = np.linspace(0, 2*np.pi, 1000)

AF = 2 * np.sin((k * d / 2) * np.cos(theta))

AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Polar plot
plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nEqual\nAmplitude & Opposite Phase (d = λ/2)\n\n")
plt.show()
```

Output



Result & Discussion

This experiment demonstrates that phase difference is a critical factor in antenna array behavior. Even with identical amplitudes and spacing, changing the phase from 0° to 180° completely alters the radiation direction. This principle is widely used in phased array antennas to electronically control beam direction without physically moving the antenna.

8. Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Unequal Amplitude and Opposite Phase ($d = \lambda/2$)

Objective

To analyze the combined effect of unequal amplitude and opposite phase on the radiation pattern of two isotropic sources.

Theory

In practical antenna systems, elements often have different excitation amplitudes due to design or losses. When two sources have unequal amplitudes and are excited with opposite phase, the array factor becomes:

$$AF(\theta) = A_1 e^{j\psi} - A_2 e^{-j\psi}$$

$$\psi = \frac{kd}{2} \cos \theta$$

Code

```
import numpy as np
import matplotlib.pyplot as plt

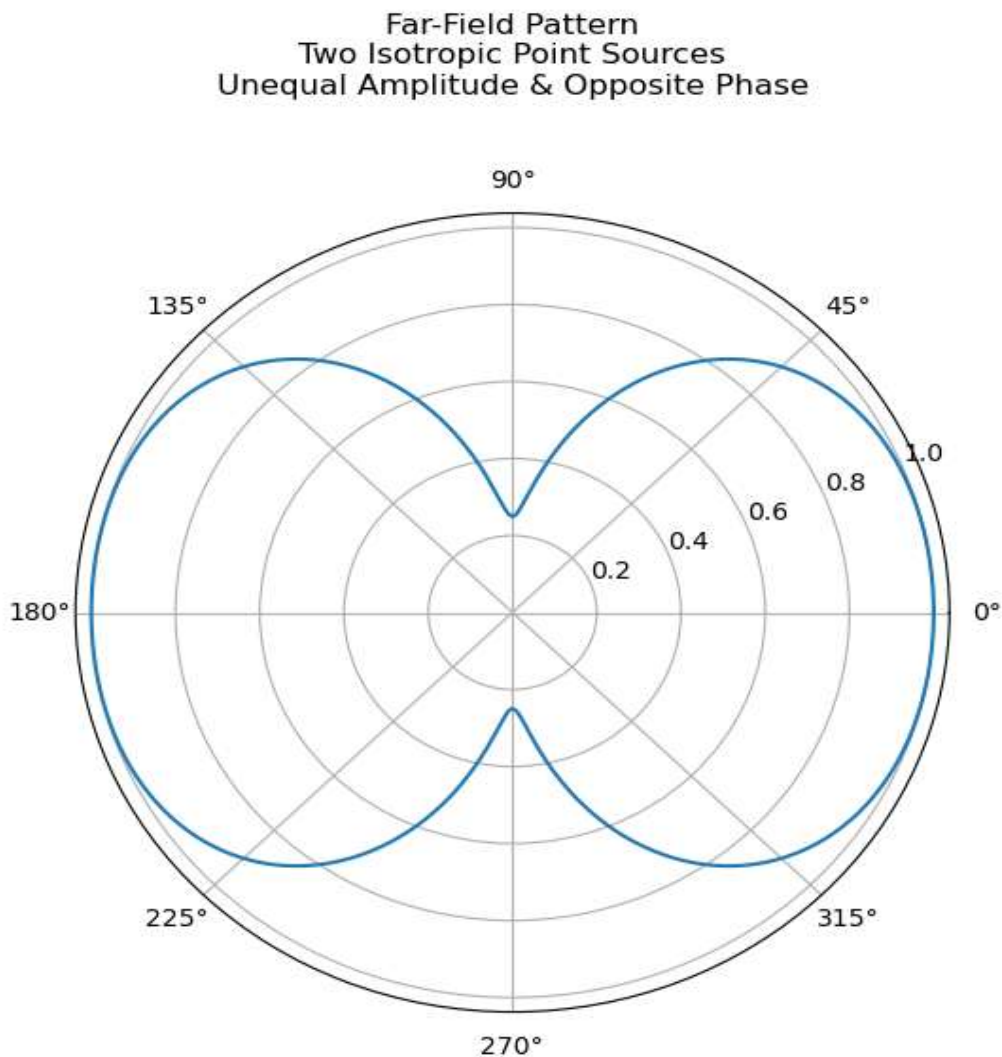
lam = 1.0          # wavelength (normalized)
d = lam / 2        # spacing
k = 2 * np.pi / lam # wave number
A1 = 1.0           # Amplitude of source 1
A2 = 0.6           # Amplitude of source 2 (UNEQUAL)

theta = np.linspace(0, 2*np.pi, 1000)
psi = (k * d / 2) * np.cos(theta)
AF = A1 * np.exp(1j * psi) - A2 * np.exp(-1j * psi)
AF_norm = np.abs(AF) / np.max(np.abs(AF))

plt.figure(figsize=(6,6))
```

```
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nUnequal\nAmplitude & Opposite Phase\n\n")
plt.show()
```

Output



Result & Discussion

This lab demonstrates that both amplitude and phase jointly control antenna radiation patterns. Unlike ideal cases, real antennas often use amplitude weighting to suppress side lobes or steer beams. This principle is fundamental in adaptive antennas, smart antennas, and beamforming systems.

9. Study and Simulation of the 2D Radiation Pattern of a Broadside 5 Element Array Antenna

Objective

To study how increasing the number of elements ($N = 5$) affects the radiation pattern of a broadside array.

Theory

A broadside array consists of multiple elements placed linearly and excited with equal amplitude and zero phase difference. The array factor is:

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jnk d \cos \theta}$$

Code

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 5 # number of elements (changed )

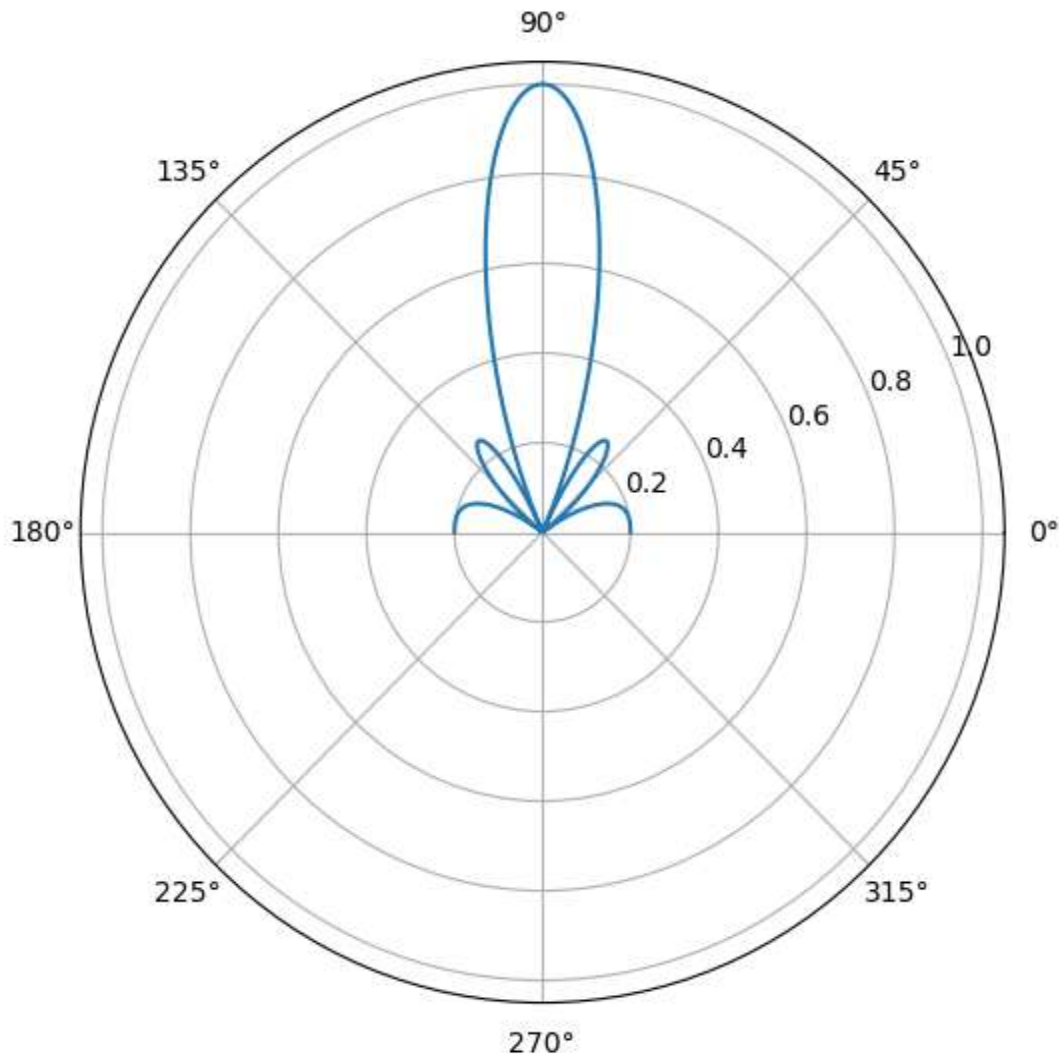
theta = np.linspace(0, np.pi, 1000)
AF = np.zeros_like(theta, dtype=complex)
for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
ax.set_title("2D Radiation Pattern (5-Element Broadside Array)\n")
plt.show()
```

Output

2D Radiation Pattern (5-Element Broadside Array)



Result & Discussion

Increasing the number of elements improves beam focusing and gain, which is desirable in communication systems. However, it also introduces side lobes, which can cause interference. This creates a design trade-off between directivity and side lobe levels, a key consideration in antenna engineering.

10. Study and Simulation of the 2D Radiation Pattern of a Broadside 7 Element Array Antenna

Objective

To analyze the radiation pattern of a 7-element broadside array and observe advanced beam shaping characteristics.

Theory

With more elements ($N=7$), the array factor becomes more complex, but follows the same principle:

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jnkd \cos \theta}$$

Code

```
import numpy as np
import matplotlib.pyplot as plt

lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 7 # number of elements (changed )

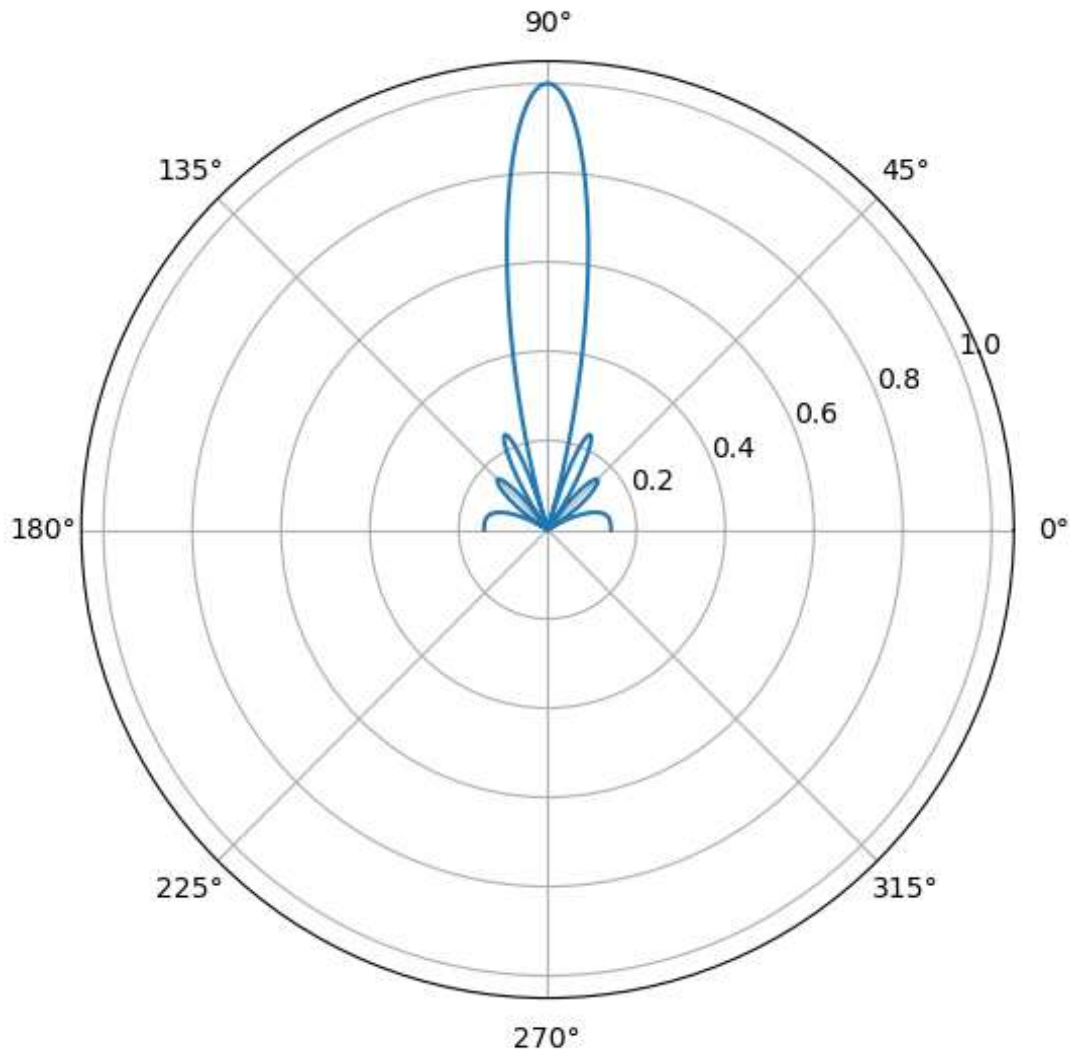
theta = np.linspace(0, np.pi, 1000)
AF = np.zeros_like(theta, dtype=complex)
for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)
ax.set_title("2D Radiation Pattern (7-Element Broadside Array)\n")
plt.show()
```


Output

2D Radiation Pattern (7-Element Broadside Array)



Result & Discussion

This experiment shows how large arrays are used in modern communication systems, including: Radar systems, Satellite communication, 5G and massive MIMO. However, increasing elements also increases system complexity and side lobes. Engineers often apply amplitude tapering techniques to reduce unwanted radiation. This lab highlights the balance between performance and practical limitations in antenna array design.